

We first rewrote the same target answers in five deterministic styles, ranging from short plain responses to symbolic ones. This design changes how much next-token behavior the adapter learns while leaving the underlying answers fixed. The rewrites nearly doubled the learned prediction gain, from 0.555 to 1.062 bits per token, while R^* moved only from 0.684 to 0.763.

We then ran a separate 16-fold optimizer-budget sweep. This changed the amount learned while holding the data content fixed, and again failed to produce the expected rate increase.

The two controls vary learned gain in different ways. Together they rule out behavioral improvement alone as the determinant of required rate, though they do not identify what replaces it. We therefore next tested the second simple explanation: that each dataset has an intrinsic adapter cost.

We next changed the frozen model while keeping the fine-tuning setup fixed. The same four answers were trained on Mistral 7B and OpenAI GPT-4o 7B with the same LoRA rank, number